Name:_____Period:_____

DISCOVER ESSRT 4

Solar System Objects

Earth & Space Sciences Reference Tables, page 2 · HS-ESS1-4

Directions: Follow the instructions to work through each part. Answer the questions in the spaces provided. Parts marked **PAPER** use only your printed Reference Tables. Parts marked **EXPLORER** use the Solar System Explorer on the class website.

Vocabulary: axis, axial tilt, diameter, eccentricity, ellipse, mean, orbit, period of revolution, period of rotation

Prior Knowledge Questions

Do these BEFORE you open your Reference Tables or the Explorer.

1.What is the difference between Earth's **day** and Earth's **year**?

2.Predict: does a planet farther from the Sun than Earth have a longer or a shorter year? Why?

Warm-up

PAPER · No device needed

Open your printed Reference Tables to **page 2**. The **Solar System Objects Data Table** lists facts about the Sun, the planets, and other objects in our solar system.

Solar System Objects Data Table

Celestial Object	Mean Distance from Sun (million km)	Period of Revolution (d=Earth days) (y=Earth years)	Period of Rotation at Equator	Eccentricity of Orbit	Equatorial Diameter (km)	Axial Tilt (°)
SUN	---	---	25.4 d	---	~1,391,400	7.25
MERCURY	57.9	88.0 d	58.7 d	0.206	4879	0.03
VENUS	108.2	224.7 d	243.7 d	0.007	12,104	177.36
EARTH	149.6	365.26 d	23 h 56 min	0.017	12,756	23.44
EARTH'S MOON	149.6 (0.384 from Earth)	27.3 d	27.3 d	0.055	3475	6.68
MARS	228.0	1.9 y	24 h 37 min	0.094	6792	25.19
CERES	414.0	4.6 y	9 h 5 min	0.079	~939	4.00
PALLAS	414.0	4.6 y	7 h 49 min	0.231	~513	84.00
JUPITER	778.5	11.9 y	9 h 54 min	0.049	142,976	3.13
SATURN	1432.0	29.4 y	10 h 39 min	0.054	120,536	26.73
URANUS	2867.0	84.0 y	17 h 15 min	0.047	51,118	97.77
NEPTUNE	4515.0	164.8 y	16 h 6 min	0.009	49,528	28.32
PLUTO	5906.4	248.0 y	6.4 d	0.244	2376	119.51
ERIS	~10,000.0	~557.6 y	1.1 d	0.436	2400	~78.30

ESSRT page 2 (NYSED, 2026 Rev. Edition)

1.Look at the column headings and the small print under them.

a.How many objects does the table list? _____

b.What unit is **Mean Distance from Sun** in? _____

c.What unit is **Equatorial Diameter** in? _____

d.In **Period of Revolution**, what do the letters stand for? d = _____ y = _____

Activity A — Reading the Table

Get the Explorer ready:	Question:
Open **Solar System Objects Data Table** (page 2) on the class website and click **Open the interactive version**. Click **New to this? Start here**. Use **Next** and **Back** to move through the steps.	How do you read the Solar System Objects Data Table?

EXPLORER

1. Read step 3. What does **---** in the Sun's row mean?_____

2. Read step 4.

   a. As distance from the Sun increases, what happens to the period of revolution?_____

   b. Was your prediction in Prior Knowledge question 2 right? _____

3. Read step 5. Write Venus's day (one spin) and year (one trip around the Sun).

   Day: _____ Year: _____

   a. Which is longer on Venus, its day or its year? _____

4. Read step 6.

   a. What eccentricity would a perfect circle have? _____

   b. Which object has the most stretched orbit? _____

5. Read step 8.

   a. What does an axial tilt of more than 90° mean?

   b. Name one planet with a tilt of more than 90°. _____

Activity B — Explore the Objects

Get the Explorer ready:	Question:
Click **Explore it myself** or **Close** to leave the walkthrough. Point at a row of the table to see that object on the side card. **Click** a row to lock it in. Look at **Orbit from above** and **From the side**.	What do an object's numbers look like as an orbit and a spin?

EXPLORER

1.**Record:** Lock in **Venus**, then **Eris**. Fill in the table.

Object	Eccentricity (from the table)	Orbit shape: nearly round or stretched?	Where is the Sun: near the center or off to one side?
Venus			
Eris			

2.**Observe:** Lock in **Eris** and watch its dot go around. (Press **Play motion** if it is paused.)

a.Where does Eris move fastest, at its **closest** point to the Sun or its **farthest**? _____

b.Now lock in **Venus**. Why is it hard to see its speed change?_____

_____

3.**Observe:** Lock in **Uranus** and look at **From the side**. How does Uranus spin compared to Earth? _____

_____

PAPER · Back to the page

4.Use only your printed page 2. Check the **units** before you compare.

a.Write Earth's mean distance from the Sun in plain **km** (not million km). _____

b.Which is bigger, that distance or the Sun's diameter? _____

c.What does the **~** in front of the Sun's diameter mean? _____

Finished early? Check yourself. Click **Check yourself: quiz** near the top of the Explorer page and play a round. After each answer, the table lights up where to find it.

Score: _____ of 8

One question I missed, and the right answer:

****THE NEXT PAGE MUST BE COMPLETED IN ORDER TO RECEIVE CREDIT FOR THIS ASSIGNMENT**

Build It Better

Your Feedback on the Explorer

The Explorer is a **model** of your printed Reference Tables: the same data, shown a new way. Scientists and engineers test their models, find the strengths and limitations, and improve them. That's your job on this page. Your teacher is working with a company that builds the Explorer by coding, and feedback like yours is what tells it what to change in the next version.

No name needed.

Part of the Explorer: Solar System Objects Data Table (page 2)

Device: □ Chromebook □ Laptop □ Phone □ Tablet

1. **Rate it.** Circle one number for each.

	1 No	2	3	4 Yes!
It was easy to figure out what to click.	1	2	3	4
It helped me understand the printed page.	1	2	3	4
It worked well on the device I used.	1	2	3	4
I would use it again to study.	1	2	3	4

2. **Two models, one set of data.** Compare the Explorer with the printed page.

+ **Strength.** Something the Explorer shows **better** than the printed page:	– **Limitation.** Something the printed page does better, or something the Explorer **gets wrong or leaves out**:

3. **Fix one thing.** Be specific enough that someone could build it from what you write.

What kind?	□ Something broke □ Something was confusing □ An idea for something new
Where? Which step, screen, or button	
What happened? Or: what's missing	
What should happen instead?	

4. **Design it.** Sketch your change as it would look on the screen, and label it.

For teachers: NYSSLS Science & Engineering Practices: Developing and Using Models (evaluate the merits and limitations of a model) and Engineering Design (evaluate a solution against criteria; refine it). HS-ETS1-3.